Chebyshev and FIR design

Answers and worked solutions · Day 20 · Thu 12 Nov · pack D12

5. Practice

From your course files

T9 Q1 (tutorial).

Hint: integrate $e^{j\omega n}$ from $-\omega_c$ to ω_c.

Inverse DTFT of $H(\omega)$:

$$h(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{j\omega n}\,d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega n}\,d\omega = \frac{1}{j2\pi n}\left[e^{j\omega n}\right]_{-\omega_c}^{\omega_c} = \frac{1}{\pi n}\sin(\omega_c n)$$

$$h(n) = 2f_c\,\mathrm{sinc}(\omega_c n), \qquad \mathrm{sinc}\,x = \begin{cases} 1, & x = 0 \\ \dfrac{\sin x}{x}, & \text{otherwise} \end{cases}$$

where $f_c = \omega_c/2\pi$. (The course sinc is $\sin x/x$; MATLAB's is $\sin(\pi x)/(\pi x)$.)

The ideal LPF is not realisable in practice for two reasons:

- $h(n)$ is not causal, because $h(n) \neq 0$ for $n < 0$;
- $h(n)$ is of infinite duration.

T9 Q7 (tutorial).

Hint: Playbook A. Prewarp first, and use $\cosh^{-1}$ (section 1.2).

Edges. $\omega_p = \dfrac{2\pi\cdot 2}{40} = \dfrac{\pi}{10}$, $\omega_s = \dfrac{2\pi\cdot 4}{40} = \dfrac{\pi}{5}$. Let $T = 1$ and prewarp: $\Omega_p = 2\tan\frac{\pi}{20} = 0.3168$, $\Omega_s = 2\tan\frac{\pi}{10} = 0.6498$.

Linear scale. $\delta_p = 10^{-1/20} = 0.8913$, $\delta_s = 10^{-50/20} = 0.0032$.

$$\epsilon = \sqrt{\frac{1}{\delta_p^2} - 1} = 0.5087$$

$$N = \frac{\cosh^{-1}\left(\dfrac{1}{\epsilon}\,\dfrac{\sqrt{1-\delta_s^2}}{\delta_s}\right)}{\cosh^{-1}\left(\dfrac{\Omega_s}{\Omega_p}\right)} = 5.2851 \;\Rightarrow\; N = 6.$$

> **NOTE:** the official solution prints "$\cos^{-1}$" here. Both arguments are greater than 1, so it has to be $\cosh^{-1}$, and 5.2851 only comes out with $\cosh^{-1}$. With unrounded values, $\epsilon = 0.50885$ and $N = 5.2929$. The official 5.2851 comes from using the rounded values inside the formula ($\delta_s = 0.0032$, $\epsilon = 0.5087$ and $\Omega_s/\Omega_p = 0.6498/0.3168$). Both round up to $N = 6$, so the course answer stands.

T9 Q8 (tutorial).

Hint: the ends differ in sign, so antisymmetric. M odd, so the middle sample is 0.

The first and last samples differ, so $h(n)$ cannot be symmetric; it must be antisymmetric. $M = 5$ is odd, so the mid-point sample is $h\left(\frac{M-1}{2}\right) = h(2) = 0$:

$$h(n) = \{\underset{\uparrow}{1}, 2, 0, -2, -1\}$$

$$H(\omega) = 1 + 2e^{-j\omega} - 2e^{-j3\omega} - e^{-j4\omega} = e^{-j2\omega}\left(e^{j2\omega} - e^{-j2\omega}\right) + 2e^{-j2\omega}\left(e^{j\omega} - e^{-j\omega}\right) = 2(\sin 2\omega + 2\sin\omega)\, e^{-j(2\omega - \pi/2)}$$

The phase response is linear:

$$\theta(\omega) = \begin{cases} -2\omega + \pi/2, & \sin 2\omega + 2\sin\omega \geq 0 \\ -2\omega - \pi/2, & \text{otherwise.} \end{cases}$$

T9 Q9 (tutorial).

Hint: same steps as Example 3, with $\omega_c = \pi/6$. Hanning $w = \{0, 0.5, 1, 0.5, 0\}$.

(a)

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi/6}^{\pi/6} e^{j\omega n}\, d\omega = \frac{1}{6}\,\mathrm{sinc}\left(\frac{\pi n}{6}\right)$$

(b) Shift by $(M-1)/2 = 2$ and window:

$$h(n) = h_d\left(n - \tfrac{M-1}{2}\right)w(n) = \frac{1}{6}\,\mathrm{sinc}\left(\frac{\pi}{6}(n-2)\right)w(n), \qquad w(n) = 1,\ 0 \leq n \leq 4.$$

$$h(n) = \left\{\tfrac{1}{6}\mathrm{sinc}\left(-\tfrac{\pi}{3}\right), \tfrac{1}{6}\mathrm{sinc}\left(-\tfrac{\pi}{6}\right), \tfrac{1}{6}\mathrm{sinc}(0), \tfrac{1}{6}\mathrm{sinc}\left(\tfrac{\pi}{6}\right), \tfrac{1}{6}\mathrm{sinc}\left(\tfrac{\pi}{3}\right)\right\} = \{\underset{\uparrow}{0.1378}, 0.1592, 0.1667, 0.1592, 0.1378\}$$

(c) Hanning window $w(n) = \frac{1}{2}(1 - \cos(\pi n/2))$ for $0 \leq n \leq 4$, that is $\{0, 0.5, 1, 0.5, 0\}$:

$$h(n) = \{0.1378, 0.1592, 0.1667, 0.1592, 0.1378\} \times \{0, 0.5, 1, 0.5, 0\} = \{\underset{\uparrow}{0}, 0.0796, 0.1667, 0.0796, 0\}$$

T9 Q4 (tutorial; only if not done in D11).

Hint: only the frequencies are warped. The magnitude limits stay the same.

Bilinear transformation requires frequency prewarping, $\Omega = \frac{2}{T}\tan\frac{\omega}{2}$:

$$0.8 \le |H(\Omega)| \le 1, \quad 0 \le \Omega \le \frac{2}{T}\tan 0.15\pi \ \ (= 1.0191 \text{ with } T = 1)$$

$$|H(\Omega)| \le 0.1, \quad \Omega \ge \frac{2}{T}\tan 0.3\pi \ \ (= 2.7528 \text{ with } T = 1).$$

T9 Q2 (tutorial; only if not done in D11).

Hint: the official solution uses $2/T = 20$ (that is, $T = 0.1$). Map the quadratic with $b = 0.2$, $c = 9.01$.

Substitute $s = 20\,\dfrac{1 - z^{-1}}{1 + z^{-1}}$ and multiply top and bottom by $(1 + z^{-1})^2$:

- numerator: $\left[20(1 - z^{-1}) + 0.1(1 + z^{-1})\right](1 + z^{-1}) = 20.1 + 0.2z^{-1} - 19.9z^{-2}$
- denominator: $400(1 - z^{-1})^2 + 0.2 \cdot 20(1 - z^{-2}) + 9.01(1 + z^{-1})^2 = 413.01 - 781.98z^{-1} + 405.01z^{-2}$

$$H(z) = \frac{20.1 + 0.2z^{-1} - 19.9z^{-2}}{413.01 - 781.98z^{-1} + 405.01z^{-2}} = \frac{0.0487 + 0.0005z^{-1} - 0.0482z^{-2}}{1 - 1.8934z^{-1} + 0.9806z^{-2}} \quad (\text{after normalising } a_0 = 1)$$

Zeros $z_1 = -1$, $z_2 = 0.99$; poles $p_{1,2} = 0.9467 \pm j0.2905$.

T9 Q3 (tutorial; only if not done in D11).

Hint: warping makes the digital integrator fall to zero at $\omega = \pi$. Avoid inputs with DC (pole at $z = 1$).

(a)

$$H(z) = H_a(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}} = \frac{T}{2}\,\frac{1 + z^{-1}}{1 - z^{-1}}, \qquad |z| > 1 \text{ (causal system)}$$

$$y(n) - y(n-1) = \frac{T}{2}x(n) + \frac{T}{2}x(n-1).$$

(b) $H_a(\Omega) = \dfrac{1}{j\Omega} = -\dfrac{j}{\Omega}$, so $|H_a(\Omega)| = \dfrac{1}{|\Omega|}$ (infinite at $\Omega = 0$, a hyperbola either side) and $\theta(\Omega) = -\frac{\pi}{2}$ for $\Omega > 0$, $+\frac{\pi}{2}$ for $\Omega < 0$.

(c)

$$H(\omega) = \frac{T}{2}\,\frac{1 + e^{-j\omega}}{1 - e^{-j\omega}} = \frac{T}{2}\,\frac{e^{-j\omega/2}\left(e^{j\omega/2} + e^{-j\omega/2}\right)}{e^{-j\omega/2}\left(e^{j\omega/2} - e^{-j\omega/2}\right)} = -j\,\frac{T}{2}\,\frac{\cos(\omega/2)}{\sin(\omega/2)} = \frac{T}{2}\cot(\omega/2)\,e^{-j\pi/2}$$

$$|H(\omega)| = \frac{T}{2}|\cot(\omega/2)|, \quad -\pi \le \omega \le \pi; \qquad \theta(\omega) = \begin{cases} -\frac{\pi}{2}, & 0 < \omega \le \pi \\ \frac{\pi}{2}, & -\pi \le \omega < 0 \end{cases}$$

Strictly, $H(\omega)$ does not exist because the ROC $|z| > 1$ excludes the unit circle, but it can still be computed if the pole at $z = 1$ is avoided (L04 slides 12–13).

(d) For $T = 1$ ($\omega = \Omega$), $1/|\Omega|$ and $\frac{T}{2}|\cot(\omega/2)|$ match very well at all frequencies except near $\omega = \pi$, where the frequency warping of the bilinear transformation makes the digital integrator

decay to zero. The phase responses are identical.

(e) Input signals with a non-zero DC component should be avoided, because they would make the digital integrator unstable (unbounded output).

New questions

New Q1 [3 marks]

a. $\delta_p = 10^{-0.5/20} = 0.9441$ and $\delta_s = 10^{-30/20} = 0.03162$.

$\epsilon = \sqrt{1/0.9441^2 - 1} = \mathbf{0.3493}$ ($\epsilon^2 = 0.1220$).

Numerator: $\frac{\sqrt{1-0.001}}{0.3493 \times 0.03162} = 90.48$, and $\cosh^{-1}(90.48) = 5.1983$.

Denominator: $\cosh^{-1}(2) = 1.3170$.

$N = 5.1983/1.3170 = 3.947 \Rightarrow \mathbf{N = 4}$.

b. Butterworth (L10 slide 17): $N = \dfrac{\log \frac{\delta_p^2(1-\delta_s^2)}{\delta_s^2(1-\delta_p^2)}}{2\log 2} = 6.4996 \Rightarrow \mathbf{N = 7}$.

The Chebyshev filter saves three orders. scipy check: `cheb1ord(1000,2000,0.5,30,analog=True)` gives 4, and `buttord` gives 7. For reference, `ellipord` gives 3.

New Q2 [4 marks]

$\epsilon = \sqrt{10^{0.1} - 1} = 0.5088$.

$\beta = \left(\frac{\sqrt{1.2589}+1}{0.5088}\right)^{1/3} = 4.1702^{1/3} = 1.6096$.

$r_1 = \frac{1.6096^2+1}{2(1.6096)} = 1.1154$ and $r_2 = \frac{1.6096^2-1}{2(1.6096)} = 0.4942$.

$\phi_k = 90^\circ + (2k+1)30^\circ = 120^\circ, 180^\circ, 240^\circ$. The poles are:

- $p_0 = 0.4942\cos 120^\circ + j1.1154 \sin 120^\circ = \mathbf{-0.2471 + j0.9660}$
- $p_1 = \mathbf{-0.4942}$
- $p_2 = \mathbf{-0.2471 - j0.9660}$

The denominator is
$(s + 0.4942)(s^2 + 0.4942s + 0.9942) = s^3 + 0.9883s^2 + 1.2384s + 0.4913$.

N is odd, so $K = 0.4913$, and

$$H(s) = \frac{0.4913}{s^3 + 0.9883s^2 + 1.2384s + 0.4913}$$

$|H| = 1$ where $T_3(\Omega) = 4\Omega^3 - 3\Omega = 0$, which is at $\Omega = 0$ and $\Omega = \sqrt{3}/2 = 0.866$ rad/s.
$|H| = 0.8913$ (-1 dB) where $T_3^2 = 1$, which is at $\Omega = 0.5$ and $\Omega = 1$ rad/s. That gives 4 extrema across $0 \le \Omega \le 1$, the equiripple passband.

scipy check: `cheby1(3,1,1,analog=True)` and `cheb1ap(3,1)` give the same result.

New Q3 [5 marks]

a. Symmetric: $\{1,3,3,1\}$ (Type II). Antisymmetric: $\{1,3,-3,-1\}$ (Type IV).

b. $\{1,3,3,1\} = (1+z^{-1})^3$ has a triple zero at $z=-1$, so $H(\pi)=0$. It cannot pass high frequencies.

$\{1,3,-3,-1\}$: $H(0) = 1+3-3-1 = 0$, a zero at $z=1$. Factorising, $z^3+3z^2-3z-1 = (z-1)(z^2+4z+1)$, so the zeros are $z=1$, -0.2679 and -3.7321. The last two are a reciprocal pair ($0.2679 = 1/3.7321$). $|H(\pi)| = |1-3-3+1| = 4$. The response is 0 at DC and 4 at $\omega=\pi$, so the **antisymmetric** filter is the highpass candidate.

c. $\alpha = 1.5$.

$$H(\omega) = e^{-j1.5\omega}\left[(e^{j1.5\omega}-e^{-j1.5\omega}) + 3(e^{j0.5\omega}-e^{-j0.5\omega})\right] = 2\left[\sin 1.5\omega + 3\sin 0.5\omega\right]e^{-j(1.5\omega-\pi/2)}$$

The group delay is **1.5 samples**. Checked numerically: the unwrapped phase $+1.5\omega$ equals $\pi/2$ at every test frequency.

New Q4 [8 marks]

a. $h_d(n) = \dfrac{\sin(\pi n/2)}{\pi n} = \frac{1}{2}\mathrm{sinc}\left(\frac{\pi n}{2}\right)$:

- $h_d(0) = 0.5$
- $h_d(\pm 1) = 1/\pi = 0.3183$
- $h_d(\pm 2) = 0$
- $h_d(\pm 3) = -1/(3\pi) = -0.1061$

b. $\alpha = 3$, so $h(n) = h_d(n-3)$:

$$h(n) = \{-0.1061,\ 0,\ 0.3183,\ 0.5,\ 0.3183,\ 0,\ -0.1061\}.$$

c. $w(n) = 0.54 - 0.46\cos(\pi n/3) = \{0.08,\ 0.31,\ 0.77,\ 1,\ 0.77,\ 0.31,\ 0.08\}$, so

$$h(n) = \{-0.0085,\ 0,\ 0.2451,\ 0.5,\ 0.2451,\ 0,\ -0.0085\}.$$

d. $A(\omega) = h(3) + 2h(2)\cos\omega + 2h(1)\cos 2\omega + 2h(0)\cos 3\omega$.

	$A(0)$	$A(\pi/2)$	$\lvert A(\pi)\rvert$
Rectangular	$0.5 + 0.6366 - 0.2122 = 0.9244$	0.5	$0.5 - 0.6366 + 0.2122 = 0.0756$ (−22.4 dB)
Hamming	$0.5 + 0.4902 - 0.0170 = 0.9732$	0.5	$0.5 - 0.4902 + 0.0170 = 0.0268$ (−31.4 dB)

Hamming gives the better stopband: −31.4 dB versus −22.4 dB at $\omega=\pi$. Both are exactly 0.5 (−6 dB) at the cutoff.

e. Symmetric with $M = 7$ (odd), so **Type I**. The group delay is $\alpha = 3$ samples.

scipy check: `firwin(7, 0.5, window='boxcar'/'hamming', scale=False)` reproduces (b) and (c).

New Q5 [5 marks]

Check the prototype first: $|H_{LP}(\omega)| = |\cos(\omega/2)|$, which is 0.7071 at $\pi/2$. Correct.

$a = -\dfrac{\cos((\pi/2+\pi/3)/2)}{\cos((\pi/2-\pi/3)/2)} = -\dfrac{\cos 75^\circ}{\cos 15^\circ} = -0.2679$. (Exactly $a = \sqrt{3}-2$.)

Substitute $z^{-1} \to -\dfrac{z^{-1}+a}{1+az^{-1}}$:

$$H_{HP}(z) = \frac{1}{2}\left(1 - \frac{z^{-1}+a}{1+az^{-1}}\right) = \frac{(1-a)}{2}\cdot\frac{1-z^{-1}}{1+az^{-1}} = \boxed{\frac{0.6340\,(1-z^{-1})}{1-0.2679z^{-1}}}$$

Checks (freqz):

- $|H(0)| = 0$
- $|H(\pi/3)| = 0.7071$ (−3.01 dB), the same gain the prototype had at its edge
- $|H(\pi/2)| = 0.866$
- $|H(\pi)| = 0.6340 \times 2/1.2679 = 1.000$

So the filter is a highpass with its −3 dB edge at $\pi/3$.

New Q6 [6 marks]

a. $\Omega_p = 2\tan(0.1\pi) = 0.6498$ and $\Omega_s = 2\tan(0.175\pi) = 1.2256$. The ratio is 1.8860.

$\delta_p = 0.9441$ and $\delta_s = 0.01$, so $\epsilon = \mathbf{0.3493}$.

$\cosh^{-1}\left(\frac{\sqrt{1-10^{-4}}}{0.3493\times 0.01}\right) = \cosh^{-1}(286.26) = 6.3501$ and $\cosh^{-1}(1.8860) = 1.2485$.

$N = 5.086 \Rightarrow \mathbf{N = 6}$. It is only just over 5, but you still round up.

b. Butterworth needs $N = 8.92 \to 9$, which is more taps but a smoother phase. Elliptic needs $N = 4$ (scipy `ellipord`), which is the lowest order but the most nonlinear phase. Chebyshev I and II both need 6.

c. $|H| = 0$ when $T_6(\Omega_s/\Omega) = 0$, which is when $\Omega_s/\Omega = \cos\frac{(2k+1)\pi}{12}$ for $k = 0, 1, 2$. That gives $\Omega = 1.2256/\cos 15^\circ$, $1.2256/\cos 45^\circ$ and $1.2256/\cos 75^\circ$, i.e. **1.2688, 1.7333 and 4.7354 rad/s**.

These are the zeros of the equiripple stopband. All of them lie above Ω_s.

scipy check: `cheb2ord(0.2,0.35,0.5,40)` returns $N = 6$, and the `cheby2` design gives −0.50 dB at 0.2π and −40.06 dB at 0.35π.

6. Quick recall check

Question

1. Elliptic is the most efficient. Bessel has the best (linear) passband phase.
2. Even N, so $|H(0)| = \delta_p$, which is −2 dB.
3. The ends are 2 and -2, so antisymmetric. $M = 5$ is odd, so $h(2) = 0$, and $h(1) = -h(3) = 1$. $h = \{2, 1, 0, -1, -2\}$.
4. $w(n) = 0.54 - 0.46\cos(\pi n) = \{0.08,\ 1,\ 0.08\}$.
5. It always has a zero at $z = -1$, so $H(\pi) = \sum(-1)^n h(n) = 0$.